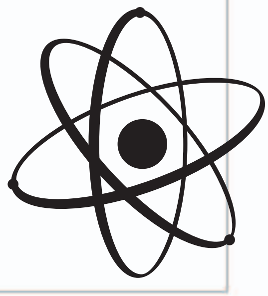
CHEMICAL ENGINEER

JOB DESCRIPTION

Chemical engineers develop the technologies that turn raw materials into useful products, such as paints, glues, textiles, and plastics. Some work in laboratories, designing new—or improving existing—products, while others specialize in developing efficient manufacturing processes—the machinery and techniques used to produce the products while meeting quality and safety standards.



SALARY

Associate engineer ★★☆☆☆
Senior engineer ★★★★☆

INDUSTRY PROFILE

Huge global industry • Rising energy costs driving innovation • Manufacturing often based in countries with lower labor and resource costs

AT A GLANCE

YOUR INTERESTS Chemistry • Mathematics • Physics • Biology • Technology • Project Management • Computing

ENTRY QUALIFICATIONS A degree in chemical, process, or biochemical engineering, as well as some practical experience, is required.

LIFESTYLE Working hours are regular in research and development, but shiftwork may be necessary in some processing and manufacturing fields.

LOCATION The work is usually based in an office, laboratory, or chemical plant. Chemical engineers may have to travel sometimes overseas to visit sites.

THE REALITIES This is a high-pressure job demanding swift problem-solving skills. Chemical engineers may be in charge of operating expensive facilities.

SKILLS GUIDE

- Good interpersonal skills to interact with a range of people across the industry.
- Problem-solving and analytical skills to manage complex projects and large budgets.
- Mathematical skills and an ability to apply scientific principles to real-world problems.
- Expertise in specialized computer software, used to process data and control production lines.
- The ability to predict and analyze the commercial results of scientific applications.
- Creativity and innovation to define manufacturing processes that make industrial products.